RAJALAKSHMI ENGINEERING COLLEGE

Praveen A 250701547-2025-CSE

P2

Started on	Sunday, 30 August 2026, 12:55 PM
State	Finished
Completed on	Sunday, 30 August 2026, 1:07 PM
Time taken	12 mins 2 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1;
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    }

    if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target);
    }

    return binarySearch(arr, mid + 1, high, target);
}

int findPair(int arr[], int left, int right, int x, int *elem1, int *elem2) {
    if (left >= right) {
        return 0;
    }

    int mid = left + (right - left) / 2;

    int complement = x - arr[mid];
    int matchIdx = binarySearch(arr, 0, right, complement);

    if (matchIdx != -1 && matchIdx != mid) {
        if (arr[mid] < arr[matchIdx]) {
            *elem1 = arr[mid];
            *elem2 = arr[matchIdx];
        } else {
            *elem1 = arr[matchIdx];
            *elem2 = arr[mid];
        }
        return 1;
    }

    if (findPair(arr, left, mid - 1, x, elem1, elem2)) {
        return 1;
    }

    return findPair(arr, mid + 1, right, x, elem1, elem2);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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